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ITreeForeCast: An integrated modeling software to simulate tree level growth and forest carbon storage

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ABSTRACT

Healthy trees in forest act as a natural carbon sink, capturing carbon. As they grow, they store carbon in their trunks, leaves and roots. Not all trees store carbon at the same rate, or in the same quantities, as it depends on a variety of biophysical and climatic factors. Although carbon estimation in trees can be complex, the precision of estimates is tightly linked to trees growth, both in diameter and height. However, the simulation of carbon uptake by forest and forest growth has each been modeled separately, and independently at differing levels of detail and spatial resolution. In this paper, we introduce ITreeForeCast, a simulation model combining the two types of modeling on a unified platform, enabling the investigation of impacts of management strategies on carbon sequestration and wood products. ITreeForeCast is a user-extendable framework that offers new opportunities to model, simulate, and visualize the dynamics of individual trees in a forest, simulate management strategies over time, and carbon uptake.

1. Introduction

Forests play a crucial role as carbon sinks, absorbing more carbon dioxide (CO₂) from the atmosphere than they emit (Nunes et al., 2020). Forests can store carbon for long periods of time, going decades or even longer, depending on forest management practices, natural disturbances (fires, pests, diseases), and human activities (deforestation, logging) (Jandl et al., 2007). Carbon sequestration also offers an economic opportunity, with the potential use of forests in voluntary carbon markets. Voluntary carbon markets allow businesses, individuals, and organizations to offset their carbon emissions by purchasing carbon credits (Bayon et al., 2012). This encourages efforts from entities to help reach their emission goals and contributing as a collective to reduce emissions, and present opportunities for sustainable forest management, biodiversity conservation, and community development. Strengthening carbon markets affirm the commitments to sustainable land use, and to the conservation, protection, sustainable management, and restoration of forests, through finance and investments. The voluntary carbon market provides an opportunity to invest in tropical forests management and the people who protect them (Langer & Seymour, 2023). In 2022, this market raised over \$1.3 billion globally for all types of credits, helping mitigate approximately 173 Mton of carbon emissions (Langer &

Seymour, 2023).

When examining forest biomass, it is important to investigate conditions of growth and management strategies implication to maintain a healthy forest. The management and harvesting of forest biomass, if done sustainably and effectively, can ensure resource availability, and forest ecosystem's health and resilience, maintaining its role in carbon sequestration and the overall carbon cycle (Ruiz-Peinado et al., 2017). Resource scarcity reduces the vigor of the trees and can increase susceptibility to attacks from insects and pathogens, limiting trees' ability to survive, and ultimately, to store carbon (Soong et al., 2020). However, trees store carbon at different rates, or in different quantities. It primarily varies based on a tree location along with its age, species physiology, and management. Effective management ensures the maintenance and enhancement of forest carbon sinks and helps sustain high levels of carbon stored. According to the Society of American Foresters (SAF), forest management positively contributes to (1) mitigating greenhouse gas (GHG) emissions through sequestering carbon in resilient, well-managed forests (including both trees and soil), and (2) adapting to patterns that reduces the risk of wildfire and other climate-driven disturbance (SAF, 2022). These dynamics are complex and understanding them to design adequate management now and, in the future, require advanced simulation models.

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Forest growth simulators (FGSs) have been heavily used to inform sustainable forest management strategies, at various spatial and temporal scales (Antón-Fernández & Astrup, 2022). FGSs are often divided into 2 categories, namely stand-level/landscape and individual-tree simulators (ITSs) (Nabhani & Sjølie, 2022). Stand/Landscape level models simulate changes in forest landscapes across broad spatiotemporal scales (forest stand to landscape). Two such models are the Forest Vegetation Simulator (FVS) and the 3-PG Forest Growth Model. FVS for instance, was developed by the U.S Department of Agriculture Forest Service and is used for predicting forest stand dynamics in the United States (Dixon, 2002), while the 3-PG model is a process based model, tracking changes in forest stands based on the interaction of light, productivity, water allocation and mortality (Trotsiuk et al., 2020). Generally, landscape level models are generally less data-intensive, easier to calibrate and require less computational power. Also, because of their lower resolution/granularity, they often lose information by generalizing forest ecological processes and patterns, and not capturing trees heterogeneity (Perin et al., 2021). ITSs are generally more demanding, when it comes to data and computational power, given their higher resolution. MASSIMO (Stadelmann et al., 2019), SiTree (Antón-Fernández & Astrup, 2022), are few examples. These models simulate individual tree growth, management, and forest dynamics, without accounting for carbon storage in forest ecosystems.

Carbon storage/sequestration has not been a primary focus of forest growth models and is usually modeled independently from forest growth, and at differing levels of detail and spatial resolution. Carbon content is generally estimated using tree biomass, which represents the proportion of carbon in plant dry matter (Fang et al., 2001). Ren et al. (2011) estimates the amount of biomass in quantity (mass) by unit of area, using a regression function $B = a \cdot V + b$, where B is the biomass per hectare (Mg/ha), V is the volume of trees/timber per hectare (m³/ha), and a and b are constants for a particular forest type. The increase in volume of forests and tree density are also used to predict the biomass growth (Qiu et al., 2020; Seely et al., 2002). Kurz et al. (2009) model carbon stocks and carbon stock changes resulting from Land Use, Land Use Change and Forestry (LULUCF) representing both stand- and landscape-level forest dynamics. Remote sensing has also been used to estimate forest biomass (Potter et al., 2008; Powell et al., 2010; Turner et al., 2004). This approach consists in using satellite data to produce spatially explicit information on carbon storage and derive trajectories of aboveground forest biomass for a variety of locations across the globe. LiDAR or Satellite imagery can help assess the overall health and composition of forests, identify land cover changes, provide accurate information on tree height, canopy structure, and help to estimate biomass. This approach is even more relevant when it comes to investigate large areas, as satellite data can produce map of recent forest disturbance and regrowth processes (Goward et al., 2008). Despite their relevance, these approaches overlook critical details. Like stand/landscape level FGS, currently used carbon models work at the forest level, not capturing trees heterogeneity. Species and age variability for instance can impact the accuracy of biomass estimation models. Remote sensing data also present similar limitations in spatial resolution, making it challenging to capture fine-scale details. Small-scale variations at the individual tree level may be missed.

We present ITreeForeCast, a simulation tool written in Python version 3.10.11. Unlike other ITSs, the tool integrates the simulation of carbon uptake by forest and forest growth, at the individual tree level, on a yearly basis, to examine the impact of management strategies on carbon storage and estimate the mix of products that can be obtained from these management strategies. This integration helps ensure the coherence of data and preserves the logical consistency in tree growth simulations. This capability is especially pertinent in the context of carbon storage, where the aim is to monitor the growth and height of trees in relation to various management strategies. The model is scalable, and generic enough to perform analysis over multiple geographic areas. Although TreeSim, a model proposed by Nabhani and Sjølie

(2022) offers similar features, it does not capture competition, and no insights on economic outcomes via wood products. Competition, which captures the level of availability of resources for trees, and degree of crowding by neighboring trees, is an important determinant of individual- and stand-scale growth (Das, 2012; Ford et al., 2017). ITreeForeCast offers therefore a more coherent representation of the complex socio-natural environment, capturing system feedback effects, and leading to more realistic and accurate predictions. These complex interactions between variables, when not represented on an integrated platform, would make it challenging to project carbon stocks and forest dynamics separately, raising issues of consistency in assumptions, streamlined workflow (reducing the need for separate runs), and inconsistencies, given difference in spatial scales, as noted earlier.

2. Material and methods

2.1. Study area

Testing was conducted on the Inland Empire (IE) variant. The IE variant covers forest areas in the panhandle region of Idaho, western Montana, and eastern Washington (Fig. 1). The study area is situated in the Selkirk Mountains, located in Northern Idaho, and falls within the boundaries of the Kaniksu National Forest. This region is part of the M333 Northern Rocky Mountain Forest Steppe—Coniferous Forest—Alpine Meadow Province, characterized by severe winters (Bailey, 1995). Within this province, the average temperature during the coldest month falls below 0 °C, while the warmest month typically does not exceed 22 °C. Although summers can be hot with cooler nights, the overall climatic conditions are defined by a yearly precipitation range of 510–1,020 mm, predominantly occurring in the fall, winter, and spring months. The summers are characteristically dry, influenced by westerly air masses that introduce the arid climate of the Pacific coast across the area. This results in a pronounced climatic gradient, varying distinctly from north to south and east to west. Winter brings substantial snowfall, yet permanent snowfields and glaciers are confined to relatively small areas. Predominant soils of the region consist of Inceptisols, which are generally cool and moist in nature. The vegetation in the area is predominantly composed of mixed forests featuring both evergreen and deciduous species. Table S1 (in supporting document Appendix 1) details the dominant species present within the project area. Notably, the study site has a history devoid of recorded wildfires, as per the available historical data (Center, 2020).

2.2. Software characteristics

ITreeForeCast uses an object-oriented design (OOD) which views and describes the system as a collection of interacting objects (Détienne, 1997), via the implementation of the object-oriented programming (OOP) paradigm. The key benefits of OOD includes modularity (denoting the decomposition of the system into a set of cohesive modules that can be built and revised independently), abstraction (denoting the definition of key characteristics of objects distinguishing them from each other) and encapsulation (denoting the compartmentalization of objects to separate their interface) to help preserve and facilitate maintainability, flexibility and extensibility (Chen & Crilly, 2016; Gellersen & Gaedke, 1999). OOP binds together the data and the methods that operate on them, preventing access to this data from any other part of the code. The model is written in Python (Van Rossum & Drake, 1995) which is a general-purpose language, more versatile, well-structured with a simpler syntax, offering a larger reach for further project development. Compared to C, C++, Fortran or Java languages, which often require extra memory and time to run (Nabhani & Sjølie, 2022) and R programming, Python has a lot more libraries, facilitating model extension. The model is built on equations presented in Dixon et al. (2008). ITreeForeCast is developed using modular programming, to significantly ease the burden of software maintenance by enabling

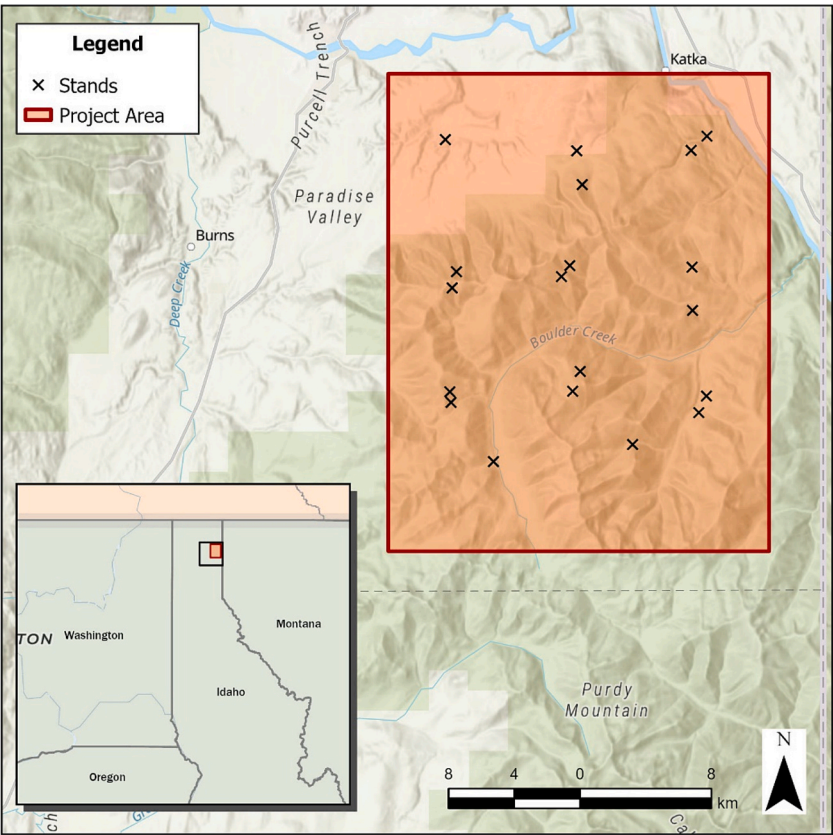


Fig. 1. Study area.

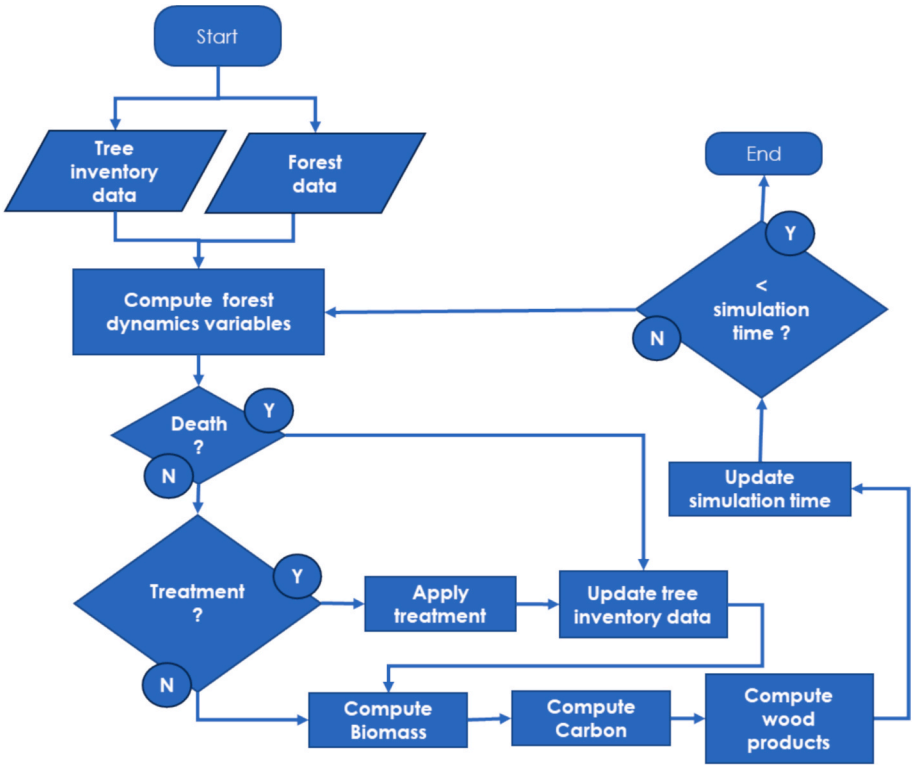


Fig. 2. Flow chart model design.

future changes and updates within isolated modules, preventing the need for extensive overhauls. This modular approach also fosters easier debugging. The inherent error isolation characteristic of modular programming streamlines the identification, isolation, and resolution of issues without disrupting other model components. Scalability is also simplified by adding or modifying modules as needed to meet evolving demands, ensuring that the application can expand seamlessly. The next section describes the software we have developed, while [section 3](#) provides an illustration of the model outputs.

2.3. Modeling methodology

The methodology presented is composed of steps taken in the software to track tree diameters at breast, height, competition factor, biomass, carbon stock, and stand composition. We used tree inventory and forest data (see [section 2.5](#)) for model parameterization. The tree records are checked for missing data and the different modules (see [section 2.4](#)) are calibrated based on the input increment data and perform the simulation. [Fig. 2](#) displays the flowchart for the simulation steps in the model.

Step 1 – Query and Read data: At the start of the simulation, there is a pre-set tree inventory, defining the initial conditions of the run. Trees info is queried based on their specific identification, stand and plot they belong to. The corresponding coefficients, all species-, forest- and location-dependent, are found and inputted for calculation.

Step 2 – Compute forest dynamics: The model computes crown competition factor (CCF) for trees and stand, tree and total basal area, tree diameter growth, and height.

Step 3 – Update inventory tree: List of trees initially inputted in the model is updated due to either death or treatment. As time progresses, trees growth is based on access to resources, which are function of the environment characteristics (topographic details, soil properties, etc.). If enough resources are available, the trees will continue to grow. The more resources there are, the more growth there will be. Low resource availability retard tree growth and may lead to death. Tree management is another reason for tree removal. In those cases, the tree inventory is updated, with potentially new trees planted. In that event, trees replanted are of the same species as the ones removed.

Step 4 – Compute biomass and carbon: As trees grow, they produce biomass, which is used to compute carbon storage. Biomass estimate is also used to compute wood products, based on proportions from tree stem, bark and branches. Using equations developed by [McTague and Stansfield \(1988\)](#), the weight of each wood product is obtained. Over the course of growth, treatments may be applied, if conditions for action are met. If not, growth continues without treatment. Treatments may enable sustained biomass collection. Once the biomass, carbon volumes and wood products are computed, the simulation time is incremented, and the cycle repeats, until the set time for simulation ends.

2.4. Software architecture and functionalities

As mentioned earlier, ITreeForeCast is designed using OOD. The model is composed of 6 modules, namely *Management*, *Carbon*, *Growth*, *Tree*, *Mortality*, and *Simulation*, predicting the dynamics and attributes of forests, including, height and diameter growth, tree mortality, biomass and carbon quantity and wood products from tree biomass ([Fig. 3](#)). Each of these modules is modeled as classes and instantiated into objects, describing the structure and static relationships between the classes identified. Classes are built by specifying the name, attributes –variables/characteristics associated with a specific object– and methods/functions –implementations of objects behavior–. Equations used in our model are extracted from [Dixon et al. \(2008\)](#) and [Ter-Mikaelian and Korzukhin \(1997\)](#). The model also contains a script, *Functions.py*, composed of multiple methods called by all modules, to not only perform all computations but also link data tables and query corresponding coefficients. It helps convert and manipulate the datasets into meaningful formats which can be used by modules during each simulation step. This script is written separately to not encumber the other modules. [Fig. 4](#) displays an illustration of one method in the script, and [Fig. 5](#) illustrates an example of data tables queries from the *Growth* module via *Functions.py* script. Equations in the growth module for instance use trees specific coefficient, which depend on different trees location and habitat classes. [Fig. 5](#) shows these coefficients. All coefficients used in the following equations and modules are in the supporting documents (see supporting document [Appendix 2](#)).

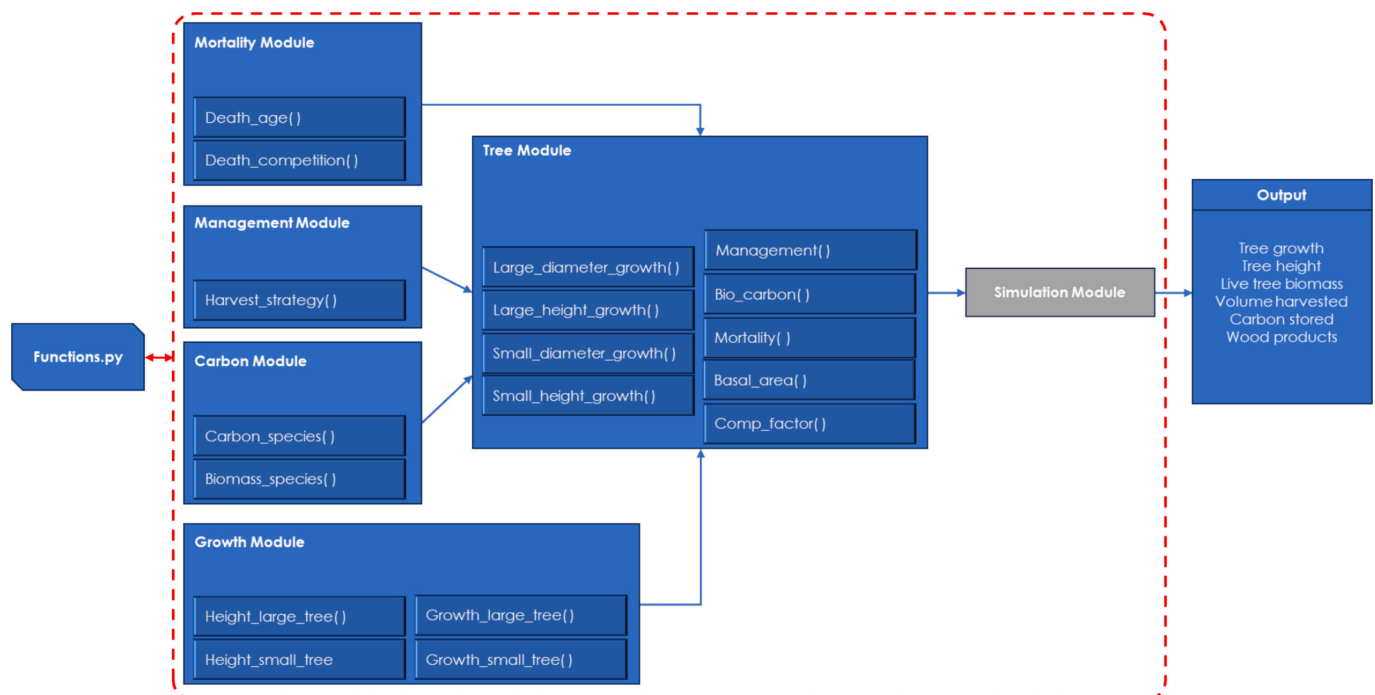
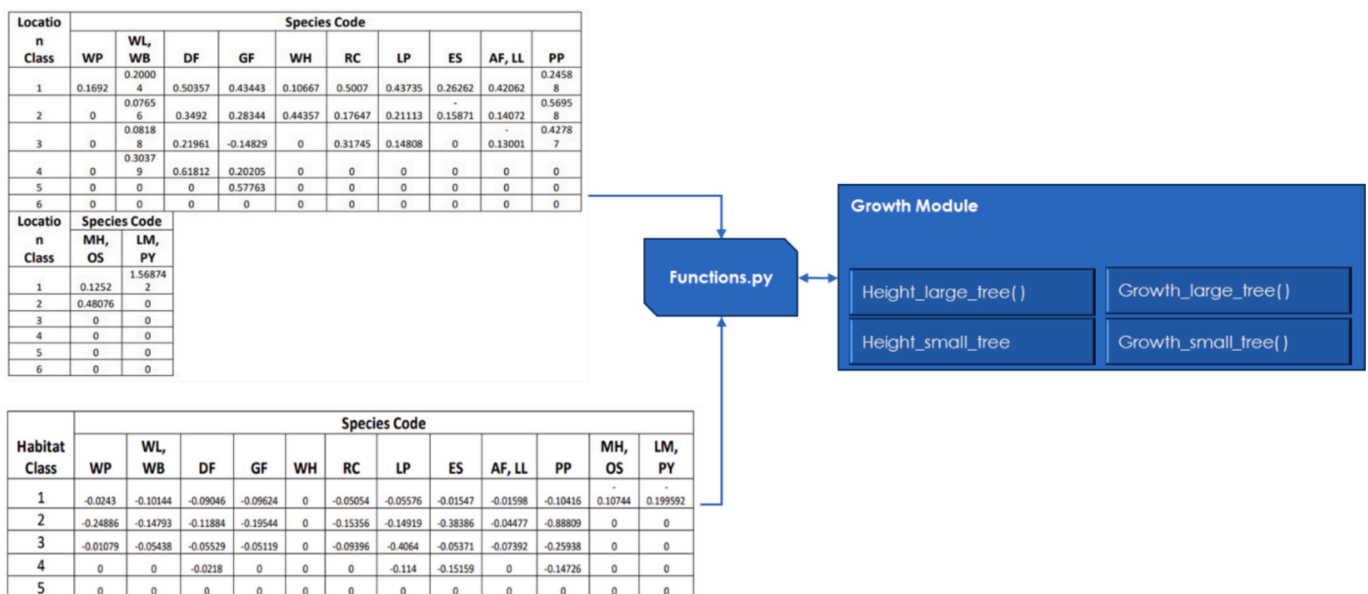


Fig. 3. Model architecture, with all classes and relationships.


```

109
110 def getSpeciesCode(data_file, tree_sp): # data_file = 321_sppCode file
111     spp_code = readData(data_file).loc[(readData(data_file)['FIA_code'] == tree_sp)][['Spp_code']]
112     return spp_code.iloc[0] # Return species code
113
114 def getHabitatCode(data_file, data_file1, data_file2, data_file3, data_file4, data_file5, standplot_id, spp_code):
115     hab_code = readData(data_file).loc[(readData(data_file)['STANDPLOT_ID'] == standplot_id)][['HABITAT']]
116     hab_code = float(hab_code.iloc[0])
117     hab_class = readData(data_file1).loc[(readData(data_file1)['Habitat Code'] == hab_code)]
118
119     for clmn in hab_class.columns:
120         if clmn == spp_code:
121             hab_class_spp = hab_class.iloc[0][clmn] #habitat class, given tree species
122             b8_coeff = readData(data_file2).loc[(readData(data_file2)['Habitat Class'] == hab_class_spp)].iloc[0][clmn]
123             b13_coeff = readData(data_file3).loc[(readData(data_file3)['Habitat Class'] == hab_class_spp)].iloc[0][clmn]
124             hab_value = readData(data_file4).loc[(readData(data_file4)['Habitat Class'] == hab_class_spp)].iloc[0][clmn]
125             b1_coeff = readData(data_file5).loc[(readData(data_file5)['Location Class'] == hab_class_spp)].iloc[0][clmn]
126             break
127     return hab_code, hab_value, hab_class_spp, b8_coeff, b13_coeff, b1_coeff
128
129
130

```

Fig. 4. Excerpt of one method in *Function.py*.Fig. 5. Example of data tables queries from the Growth module, via *Function.py* script.

2.4.1. Management module

This module captures forest management (ecosystems restoration, hazards reduction, etc.) that can potentially be applied to maintaining forest health. The effects on growth will be modeled in this module. Multiple management actions (e.g., planting, thinning, cutting) can be implemented, and at different phases of the development of the trees. The function *Harvest_strategy()* on Fig. 3 for instance, defines a harvesting strategy to implement on the trees.

2.4.2. Carbon module

This module models storage and accumulation by estimating carbon stocks, considering above ground biomass, via foliage and woody parts. Carbon storage is assumed to be 50 percent of wood biomass (Martin et al., 2021). This module also models biomass quantity for each tree, grouped by species. The biomass weight component of a tree (kg) depends on the dbh, and species-specific parameters (Ter-Mikaelian & Korzukhin, 1997). Methods *Carbon_species()* and *Biomass_species()* on Fig. 3 compute those values.

$$bio = a.D^b \quad (1)$$

Eq. (1) estimates the amount of oven-dry weight of the biomass component for each tree, with D is the tree diameter. This quantity is also dependent on coefficients a and b . These coefficients differ for each biomass component, total stem biomass (wood + bark) and foliage biomass. These coefficients are also species specific, differing for each tree species present in the stands. They can be found in (Ter-Mikaelian & Korzukhin, 1997).

2.4.3. Growth module

The *Growth* module simulates the yearly growth of tree height and diameter, across plots within a forest stand. Distinction is made separating large from small trees as they grow following a different mechanism (Dixon et al., 2008). Trees with dbh larger than 3 in. are considered large, and others, small, for certain species. Incremental growth varies depending on environmental characteristics as well as tree species. Environmental factors include slope, aspect, and elevation. Species-specific coefficients are also used for modeling growth. Tree diameter growth is impacted by competition factors, basal area, and habitat. Tree height is modeled based on habitat, location as well as environmental factors (site index, slope, and aspect).

$$htg = \exp(LOC + HAB + SPP + c1 \cdot \ln(HT) + c2 \cdot CCF + c3 \cdot BAL/100 + 0.22157 \cdot SL \cdot \cos(ASP) - 0.12432 \cdot SL \cdot \sin(ASP) - 0.10987 \cdot SL) \quad (2)$$

$$htg = \exp(HAB + b0 + (b1 \cdot HT^2) + (b2 \cdot \ln(dbh)) + (b3 \cdot \ln(HT)) + (b4 \cdot \ln(DG))) + .4809 \quad (3)$$

$$grw = c1 \cdot (HT - 4.5)^{c2} + DADJor(HT - 4.5) \cdot 10 / (SI - 4.5) \quad (4)$$

$$grw = b1 + (b2 \cdot EL) + (b3 \cdot EL^2) + (b4 \cdot \sin(ASP) \cdot SL) + (b5 \cdot \cos(ASP) \cdot SL) + (b6 \cdot SL) + (b7 \cdot SL^2) + (b8 \cdot CCF/100) + (b9 \cdot \ln(dbh)) + (b10 \cdot BAL/100) + (b11 \cdot CR) + (b12 \cdot CR^2) + (b13 \cdot dbh) + (b14 \cdot BAL / (100 \cdot (\ln(dbh + 1.0)))) + (b15 \cdot SI) + HAB \quad (5)$$

Both equations Eq. (2) and Eq. (3) calculate tree height increment – *htg*. Eq. (2) is applied to small trees, while equation Eq. (3) was used for larger trees. These height increments depend on location-specific coefficients (*LOC*), habitat type (*HAB*), tree species (*SPP*), stand crown competition factor (*CCF*), total basal area (*BAL*), slope (*SL*), aspect (*ASP*), height (*HT*) at the beginning of the cycle, and periodic increment in inside bark diameter (*DG*). *c1*, *c2*, *c3*, *b1*, ..., *b4* are coefficients specific to tree species.

Both equations Eq. (4) and Eq. (5) calculate tree growth increment – *grw*. Just like for the trees' height, growth equations are also linked to tree sizes (Eq. (4) for small trees and Eq. (5) for large trees). Eq. (4) shows 2 expressions applied based on tree species. The diameter growth increments are function of *dbh*, *HAB*, *SL*, *BAL*, elevation (*EL*), tree's live crown ratio (*CR*), stand crown competition factor (*CCF*), adjustment factor to correct for bias relative tree size and stand density (*DADJ*), species site index (*SI*), tree height (*HT*), being calculated in the previous equations. *DADJ* is a function of average height of the 40 largest diameter trees and *CCF*. *c1*, *c2*, *b1*, ..., *b15* are also coefficients specific to tree species.

2.4.4. Mortality module

This module captures the mortality mechanism, accounting for competition, randomness (capturing the effects of insects and any other factors), and age. The dynamics of life expectancy for each species as well as resource needs are modeled to estimate when and what tree is bound to die.

$$mort_rate = \frac{\sum_{sp} tr}{\sum Tr} \quad (6)$$

In this module, trees die due to old age and stress. Life expectancy varies by tree species, so trees die at different rates. Access to resources for growth is also captured, representing stress or competition. A tree size and competition from other trees are affecting and affected by the growth and mortality of another tree. As growth shapes tree size, size affects both the competition to the surrounding trees and the growth of the tree in the next period. In our case, this dynamic is illustrated by the ratio of diameter growth increment to diameter. Eq. (6) computes the mortality rate by tree species, tree *tr* of a given species *sp*, and *Tr* the total number of trees. The rate is the ratio of the number of dead trees over the number of trees at the initial time.

2.4.5. Tree module

Individual trees in the forest are modeled here. This module calls methods from *Management*, *Carbon*, *Growth* and *Mortality*, modeling a given management strategy, computing biomass weight and carbon, computing height and dbh growth, and modeling mortality, respectively. Depending on the characteristics (e.g., age, species, dbh or height), each tree growth dynamics is simulated to provide outputs. Forest regeneration is also captured here as human-assisted regeneration

or reforestation. The objective of this forest management is to increase forest cover, and/or species composition, following treatments involving tree removal (Jones et al., 2022). New trees planted are assumed to be the same species as trees removed.

2.4.6. The simulation module

This module performs the execution of the model, typically initializing objects, running the simulation loop, computing and collecting results. The simulation loop advances the simulation in discrete time steps, say each year. Each iteration updates the state of objects, here tree, simulates events, and update results. Events are what alter the state of objects, for instance, the increase in growth of a tree. Events are created by calling methods defined as tree attributes and modeled in other modules.

2.5. Data input

The datasets required to run the simulation in the model are tree-level, plot-level data, as well as forest environmental data. Tree-level dataset encompasses tree ID, crown ratio, height, dbh, species, position (latitude and longitude), stand ID and stand plot ID. Each tree can therefore be tracked and monitored throughout the simulation. Plot-level data encompasses slope, elevation, aspect, site index, forest type, stand ID and stand plot ID. Environmental data encompasses all species-, forest- and location-dependent factors/coefficients. Location-dependent coefficients, for instance, specify where on the US national forest system the stand/plot is located. Each of these locations has different habitats, identified by a code, which corresponds to different coefficients, based on tree species. Data used in this paper are specific to the species present in the study area, and obtained from (Dixon et al., 2008). It is, however, important to note that the software can be used for any species and forests present in the US. Users may need to use appropriate inventory data for model use.

3. Results

This model is intricately parameterized using Forest Inventory and Analysis (FIA) inventory data, providing detailed insights into the composition and characteristics of the forest stand. The model explores the forest growth and management dynamics and their repercussions on biomass and carbon storage.

3.1. Testing treatment simulation

The treatment is to remove small and shade-tolerant trees. This treatment can help in reducing competition for light, water, and nutrients, allowing more space and resources for the growth of larger, commercially valuable trees. It could also be carried out in areas prone to wildfire or outbreaks of pests and disease. This treatment would focus upon removing species' trees, with less than 6-inch diameter, "thinning from below", and more than 12-inch diameter, "thinning from above", every five years. These species include Western hemlock, Western red-cedar, Grand fir, Subalpine fir, Douglas-fir, Engelmann spruce, White-bark pine, and Rocky Mountain maple (see Table S1 in supporting document Appendix 1). By removing these trees, the forest's fire resilience can be enhanced, and long-term carbon storage is increased (Loudermilk et al., 2017). From a management perspective, this approach may help maintain the ecological balance of the forest, ensuring that it remains sustainable and productive. Commercially, the wood from the thinned trees, although smaller, can be utilized in several ways. For instance, it could be used for pulpwood, small wood crafts, or as biomass fuel, providing a source of renewable energy. Additionally, as the remaining trees grow large and healthier, they become more valuable for future harvesting.

3.2. Results and Discussion

The applied treatment was to test the functionality of our model and showcase its capabilities. The simulation was run over 100 years, and the treatment was applied after 3 years. Trees harvested and/or dead were not replaced, so forest regeneration was not considered in this scenario. Fig. 6 displays the average diameter growth trends by species. Species definition can be found in the supporting documents (see Table S2 in supporting document Appendix 1). An increasing trend indicates that smaller trees are being removed or died, leaving out more mature ones, with larger diameters, while a decreasing trend indicates otherwise. In the case of *thinning from below* treatment, we observe larger diameter average values, across tree species.

Fig. 7 displays a sample of trees (randomly selected for visualization purposes) carbon quantity, dynamically changing as they grow or die. All dead trees are assumed to also be harvested, in addition to the targeted trees and species. Although some dead trees would need to be left for wildlife habitat or other ecological reasons, this assumption is made for simplification and constitute an avenue for future improvement. Fig. 8 shows the cumulative volume of trees harvested throughout the simulation, for both *thinning from above*, and *thinning from below* treatments. The harvested volume can be harnessed, and converted into long-lasting wood products (e.g., lumber, furniture) or paper products (e.g. paper, cardboard) rather than being left to decay or burnt. When wood products are used in construction or other applications, they continue to store carbon, effectively extending the carbon sequestration associated with the original tree (Androff, 2021). Over the course of the simulation, the volume of woody material from *thinning from above* treatment overtakes the one in the other treatment. This may be due to a large number of small trees being harvested early on. As time goes on, trees not being harvested can grow and later on, being harvested in the *thinning from above* treatment.

When forest trees are harvested, the wood can be used for various purposes depending on the quality, size, and characteristics of the wood. The applications considered in the model are solid wood, pulpwood, and energy. Solid wood refers to timber that is cut into logs, lumber, or beams and used in construction, furniture, and other products where the structural integrity of the wood is maintained. It is usually higher-quality wood, coming from the main trunk of larger, mature trees. Pulpwood refers to smaller or lower-quality wood that is chipped and used primarily to make paper products. It comes from smaller or younger trees. Finally, energy wood refers to wood harvested for use as fuel to generate heat, electricity, or biofuels. It comes from lower-grade wood that is not suitable for solid wood or pulpwood applications. Figs. 9, 10 and 11 display the quantity of wood harvested, by tree species, for solid wood, pulp wood and energy wood, respectively, for the different treatments. We can observe in Fig. 9a an important quantity of Western Larch (WL) and Douglas Fir (DF), known for its strength, and

durability for furniture and construction purposes (Forest Services, 1984b). Results in Fig. 9b show high quantity of Lodgepole pine (LP), from harvesting small trees. The wood is relatively light in weight, soft, weak in bending, low in shock resistance, and can be used for interior trim, cabinetry, and light construction framing (Forest Services, 1984a). Results in Fig. 10 indicate a much lower quantity of woody material, compared to the solid wood outputs, with a maximum value of about 6000 kg of wood across the simulation. The species harvested the most for pulp wood application are LP and Grand Fir (GF) with some Western Redcedar (RC) trees. Those species are good source of wood for pulp and paper because as they yield strong, and high-quality paper (Procter & WM, 1976).

In Fig. 11, we can observe the amount of wood for energy application. Engelmann spruce (ES), LP, and Subalpine Fir (AF) species are the most harvested. The high quantity in year 5 in Fig. 11a indicates a good number of large trees being harvested, similarly to Fig. 9a. Except for the 1st time of harvest, the overall quantity of wood material for energy use is about the same as the quantity for pulp paper, and considerably less than for the solid wood application. This implies the presence of predominantly larger trees, with higher-quality wood.

With frequent harvesting, forest productivity can be promoted, and regeneration enhanced. Individual trees require water, sunlight, and nutrients to survive and grow. Given that these resources are limited, when too many trees compete with each other for these resources, some will be more prone to attacks from insects and pathogens. Fig. 12 shows the competition factor on the stand, with a descending trend. As small trees are cut off, trees left standing post-harvest potentially have better access to vital nutrients and space, allowing them to grow faster, more productively, and efficiently (Albert, 2020). This can be seen in Fig. 8 at the beginning of the simulation, until around year 60, with the *thinning from below* treatment providing more woody material quantity. Fig. 13 shows the mortality rate by tree's species and expresses the percentage of number of trees dead over the simulation time. See Eq (6).

ITreeForeCast presented here is an advanced tool, capable of capturing complex dynamics between diameter growth and height, impacted by resource competitiveness. It also provides biomass and carbon as well as other products quantity estimations, which can be obtained from harvesting, including energy, pulp and solid wood products. Analysis can be conducted both at the individual tree level, at the species and stand level, and on a yearly to multiyear time basis, offering a sound advantage over other modeling platforms.

The model integrates the simulation of carbon uptake by forest and forest growth, to analyze impacts of managements on carbon stocks. The model thus captures interactions and feedback loops (e.g. changes in forest management affect carbon stocks, which in turn affect future forest health) between the different processes taking place. This structure helps to ensure the coherence of data, consistency in assumptions, and reducing potential inconsistencies which result from mismatches in

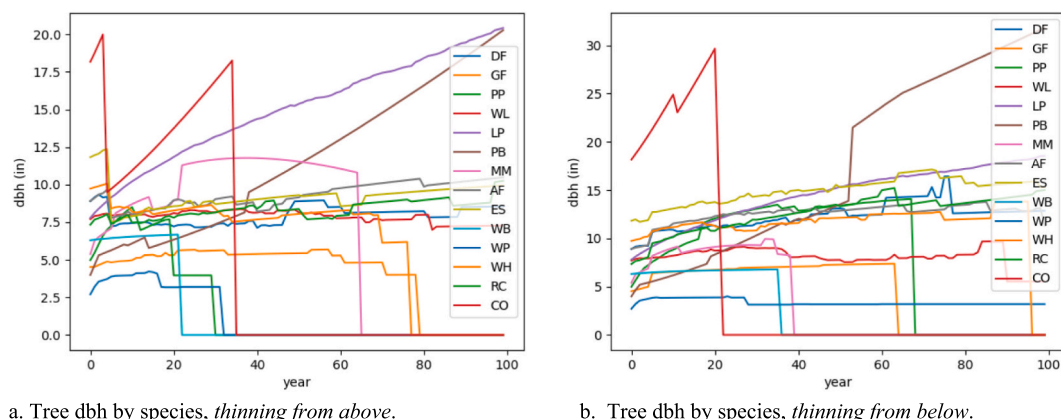


Fig. 6. Tree dbh by species.

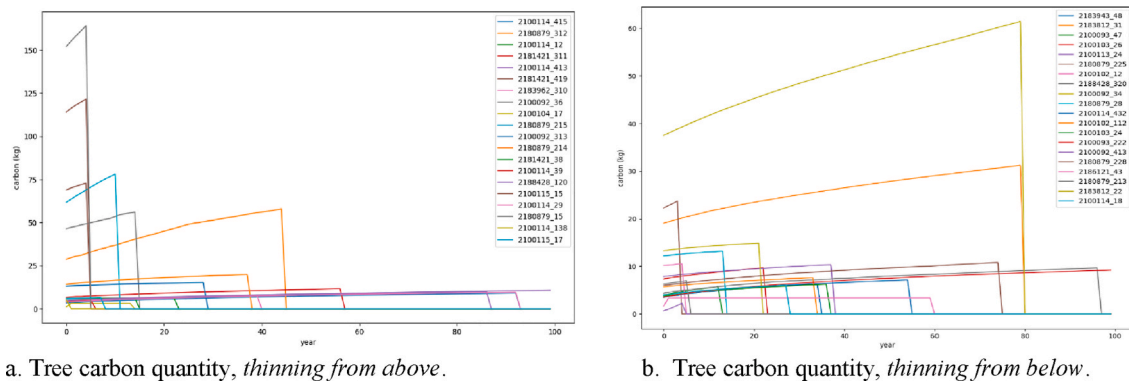


Fig. 7. Individual trees output. Graph displays a sample of trees.

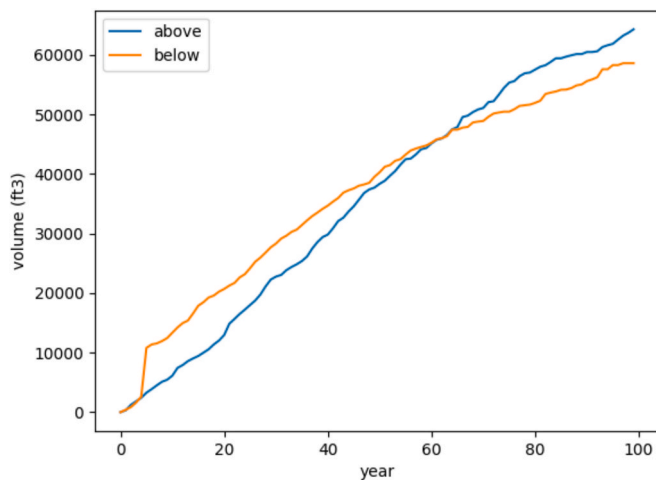


Fig. 8. Cumulative volume harvested.

spatial/temporal resolution. Beyond its structure, ITreeForeCast differentiates itself by its purpose and usefulness. Not only does it offer an analysis platform for carbon storage strategies, but it also outputs the mix of wood products emanating from treatments. This offers immense benefits by improving forest sustainability, analyzing carbon sequestration strategies, balancing economic and environmental goals, and enhancing forest resilience. This tool would enable forest managers to make informed, data-driven decisions that support economic productivity in the forestry sector, making it invaluable for both forest conservation and the timber industry. One question in our case that can be answered is: Which management strategy should be prioritized, to maximize economic/market value? Solid wood typically has the highest market value due to its use in construction and furniture. Pulpwood has a moderate value, especially in regions with large paper industries.

Energy wood tends to have the lowest value per unit of volume, although it plays an important role in energy generation and renewable fuel production. However, in terms of carbon capture, energy wood is not too useful as the wood is to be burnt.

As in every model, the question of validation is critical. The validation of outputs from forest growth models faces particular challenges due to the generally limited availability of data on long-term forest dynamics. Inter-model validations are generally performed, which typically involve comparing the outputs of one model with those of another. Additionally, model validations can be conducted for shorter periods (10–50 years) (Ngugi & Botkin, 2011) rather than validating entire simulation timespan (Porté & Bartelink, 2002). However, considering different assumptions made in models, as well as difference in scale and focus, model-to-model comparisons do not always yield useful information. Examples include SiTree, and TreeSim, which do not account for competition, or carbon dynamics, respectively. FVS primarily simulates stand-level dynamics over time, often aggregating tree-level results to provide insights into forest stand dynamics, yield, and succession. In addition, the stand growth is estimated by cycles, which are 10 years long (Dixon, 2002). ITreeForeCast model works at a finer scale, simulating individual tree growth and on a yearly basis. These differences in scope and scale make it difficult to validate/verify our model results with FVS. The validation approach used is conceptual model validity, which consists of ensuring that (1) the theories and assumptions underlying the conceptual model are correct, and (2) the model representation of the problem, and the model's structure, and mathematical relationships are logical (Sargent, 1999). Relationships and mathematical equations in the model are relevant for the intended purpose of the model, and the effects of competition, habitat, as well as species on growth are all captured. In addition, the computer program was tested, using techniques like structured walk-throughs and correctness proofs, ensuring that the program is executed as instructed. In the case of ITreeForeCast, its modular structure can play a critical role in result accuracy. With each main component of forest growth built

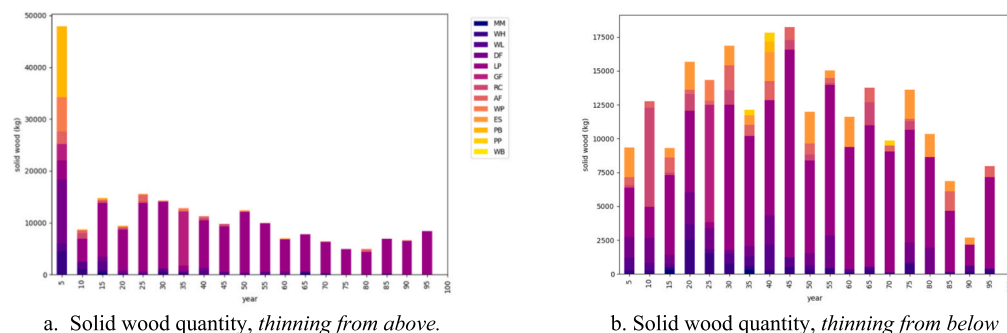


Fig. 9. Solid wood output by tree species. Colors indicate trees' species.

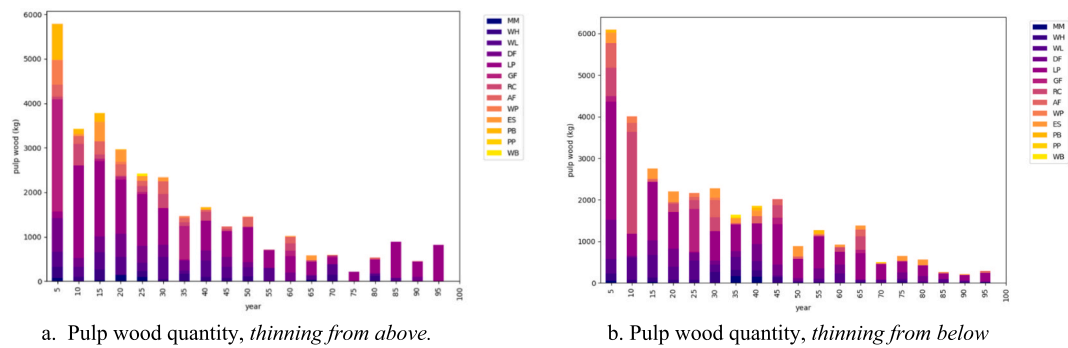


Fig. 10. Pulp wood output by tree species. Colors indicate trees' species.

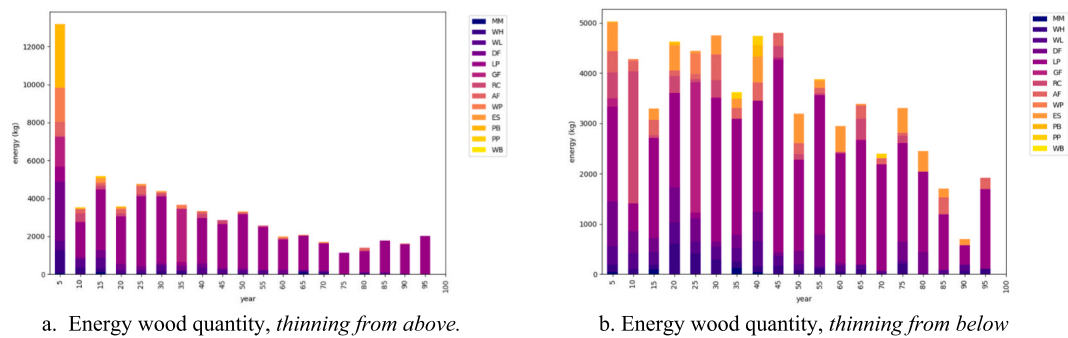


Fig. 11. Energy wood output by trees species. Colors indicate trees' species.

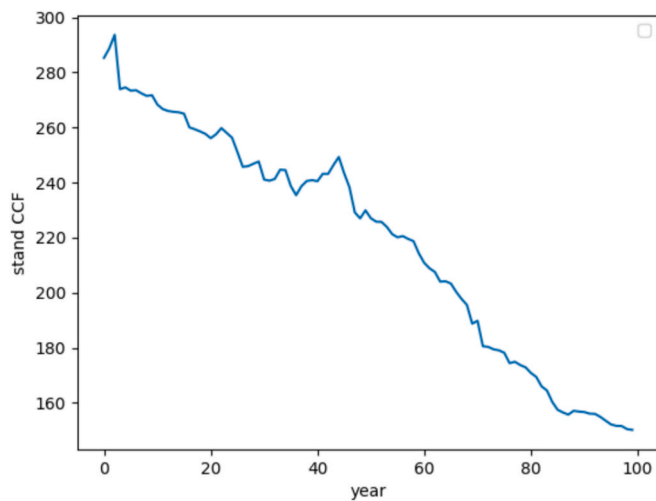


Fig. 12. Forest stand competition factor, *thinning from below*

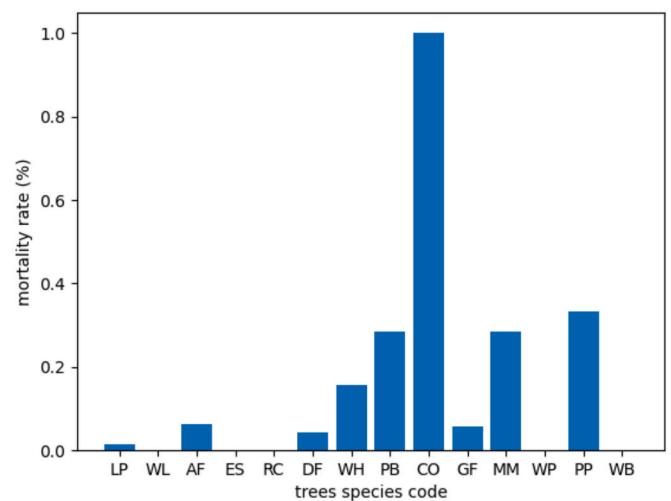


Fig. 13. Mortality rate by tree species, *thinning from below*.

independently, results verification is less speculative. Modules are independently tested, ensuring proper functioning internally, and adequate interactions with each other. Finally, we used Forest Service experts' guidance, to advise on the model structure, information data exchange and practical management strategies, which reinforce the model's results plausibility, and confidence. As we leverage real tree inventory data as input and forest dynamics equations, and ensure good model verification, results presented/obtained can be trusted to be technically sound, and reliable for our purpose. The most important is that the model can reliably reflect trends and responses under different conditions, enabling stakeholders to explore outcomes, evaluate strategies, and make informed decisions based on plausible projections.

In terms of limitation, geospatial details for tree stands are not captured. With geospatially explicit information, the model can help

identify areas within a forest stand that are most responsive to specific treatments, such as thinning or fertilization, allowing for more targeted and effective management. Another limitation is forest regeneration, which is only captured here as human-induced. Natural Regeneration such as seed dispersal enables trees to reproduce and expand their populations. Finally, the results presented here are specific to the region considered and should not be applied to another region. The model should not be used for generalization across regions, as insights offered for one region may be irrelevant to another.

4. Conclusion

We developed ITreeForeCast, a tool used to model and simulate the dynamics of forest trees growth and conduct management strategies

analysis. The model is composed of various modules to predict the dynamics and attributes of forests, including regeneration, increment, mortality, biomass, carbon and wood products. Although applied on the IE variant, its structure enables analysis on other variants. It is flexible, easily extendable and allows users to adapt the framework to specific needs, including different variants and trees inventory. As opposed to other models that simulate carbon uptake by forest and forest growth separately, this model combines the two types of modeling, offering analysis on a unified platform for forest dynamics prediction and wood products estimates. The model enables analysis at various levels, including individual trees, species, and stands, offering a more comprehensive perspective on results and their interpretation.

The outputs of the ITreeForeCast are comprehensive, including biomass volume, volume harvested, tree size (height and diameter), carbon storage and wood products. The model can help inform forest management decisions (thinning, fuel treatments, stand regeneration planning) quantifying their impacts on carbon cycling and sequestration. It can be leveraged by forestry professionals/researchers to adapt forest management practices, including timber harvesting schedules, reforestation strategies, and species selection. By simulating tree growth under different management scenarios, long-term impacts on forest and economic productivity, and carbon sequestration can be assessed. The model can also be of use to forest owners, land managers to reflect on practices that may boost the growth of higher-value timber while maintaining a stable carbon stock.

Despite the object oriented and distributed architecture of the model, facilitating extension as mentioned earlier, the model is very data intensive. The high number of species and complexities in relationships between trees and their environment require an important number of factors/coefficients. In addition, results accuracy warrants careful and extensive data consideration, which can be time consuming, and may limit access to non-advanced coders for usage. Additional features for optimizing the mix of wood products are also considered for improvement. By simulating different growth patterns and management strategies for example, the tool can determine the optimal time to harvest, based on market demands for different wood products and the growth potential of various tree species.

CRedit authorship contribution statement

Ange-Lionel Toba: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Rajiv Paudel:** Validation, Software, Methodology. **Cleve Davis:** Software, Investigation, Data curation. **Damon S. Hartley:** Writing – review & editing, Supervision, Methodology, Investigation. **Rohit Venkat Gandhi Mendadhala:** Conceptualization, Investigation, Software, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compag.2025.110495>.

Data availability

Data will be made available on request.

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